Integration by Parts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The integration by parts formula

Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$: $= xe^x - \int e^x\,dx = xe^x - e^x + C$.

2 [Foundational] • Target: The integration by parts formula

Evaluate $\int x\cos x\,dx$ using integration by parts.

Let $u = x$, $dv = \cos x\,dx$: $= x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

3 [Foundational] • Target: The integration by parts formula

Evaluate $\int \ln x\,dx$ using integration by parts (with $dv = dx$).

Let $u = \ln x$, $dv = dx$: $= x\ln x - \int x \cdot \frac{1}{x}\,dx = x\ln x - x + C$.

4 [Foundational] • Target: Repeated and definite applications

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

First pass ($u = x^2$): $x^2e^x - 2\int xe^x\,dx$. Second pass gives $\int xe^x\,dx = xe^x - e^x$. Total: $x^2e^x - 2xe^x + 2e^x + C$.

5 [Foundational] • Target: Repeated and definite applications

Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.

Let $u = x$, $dv = e^{2x}dx$: $\left[\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4}\right) - \left(0 - \frac{1}{4}\right) = \frac{e^2}{4} + \frac{1}{4}$.

6 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

$u = \ln x$, $dv = x\,dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\ln x - \int \frac{x^2}{2} \cdot \frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

7 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Evaluate $\int x^2\cos x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \cos x\,dx$: $x^2\sin x - 2\int x\sin x\,dx$. Second (on $\int x\sin x\,dx$): $= -x\cos x + \sin x$. Total: $x^2\sin x + 2x\cos x - 2\sin x + C$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The recurring-integral trick**

Evaluate $\int e^x \sin x \, dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

Let $I = \int e^x \sin x \, dx$. First pass ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x \, dx$. Second pass on the new integral ($u = \cos x$): $\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I$. So $I = e^x \sin x - e^x \cos x - I$, giving $2I = e^x(\sin x - \cos x)$, so $I = \frac{e^x(\sin x - \cos x)}{2} + C$.

9 **[Intermediate]** • **Target: Definite integrals by parts**

Evaluate $\int_0^{\pi} x \sin x \, dx$ exactly.

$u = x$, $dv = \sin x \, dx$: $[-x\cos x + \sin x]_0^{\pi} = (-\pi\cos\pi + \sin\pi) - (0) = \pi$.

10 **[Intermediate]** • **Target: Definite integrals by parts**

Evaluate $\int_1^e x^2 \ln x \, dx$ exactly.

$u = \ln x$, $dv = x^2 dx$: $\left[\frac{x^3}{3}\ln x - \frac{x^3}{9}\right]_1^e = \left(\frac{e^3}{3} - \frac{e^3}{9}\right) - \left(0 - \frac{1}{9}\right) = \frac{2e^3}{9} + \frac{1}{9} = \frac{2e^3+1}{9}$.

11 **[Intermediate]** • **Target: Reduction formulas**

Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x \, dx = x^n e^x - n\int x^{n-1} e^x \, dx$.

$du = nx^{n-1}dx$, $v = e^x$: $\int x^n e^x \, dx = x^n e^x - \int e^x \cdot nx^{n-1} \, dx = x^n e^x - n\int x^{n-1} e^x \, dx$.

12 **[Intermediate]** • **Target: Reduction formulas**

Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x \, dx$, confirming it matches an earlier result.

$\int x^2 e^x \, dx = x^2 e^x - 2\int x e^x \, dx$. Applying again: $\int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x$. Total: $x^2 e^x - 2(x e^x - e^x) = x^2 e^x - 2x e^x + 2e^x + C$, matching the earlier double-integration-by-parts result.

13 **[Intermediate]** • **Target: Integration by parts with inverse trig functions**

Evaluate $\int \tan^{-1} x \, dx$ using integration by parts with $u = \tan^{-1} x$, $dv = dx$.

$du = \frac{1}{1+x^2}dx$, $v = x$: $= x\tan^{-1} x - \int \frac{x}{1+x^2} dx = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + C$.

14 **[Intermediate]** • **Target: Integration by parts with inverse trig functions**

Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

$du = \frac{1}{\sqrt{1-x^2}}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\sin^{-1} x - \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}} dx$, which requires a further trigonometric substitution to complete.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A capstone mixed-technique problem** • **Cross-Lesson: 2.02 Continuity**

Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.

$u = x$, $dv = \cos(2x)dx$, $v = \frac{1}{2}\sin(2x)$:
$\left[\frac{x}{2}\sin(2x)\right]_0^{\pi/2} - \frac{1}{2}\int_0^{\pi/2} \sin(2x)\,dx = \left(\frac{\pi}{4}\sin\pi - 0\right) - \frac{1}{2}\left[-\frac{1}{2}\cos(2x)\right]_0^{\pi/2} = 0 - \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\right) = -\frac{1}{2}$.

16 **[Challenging]** • **Target: Visualizing an area found by integration by parts** • **Cross-Lesson: 3.01 The Derivative from First Principles**

The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.

$u = x$, $dv = e^{-x}dx$, $v = -e^{-x}$:
$[-xe^{-x}]_0^3 + \int_0^3 e^{-x}\,dx = -3e^{-3} + [-e^{-x}]_0^3 = -3e^{-3} + (1 - e^{-3}) = 1 - 4e^{-3} \approx 0.801$.

17 **[Challenging]** • **Target: Choosing between substitution and parts** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

(a) Substitution — $u = x^2$ makes $du = 2x\,dx$ appear directly, no product rule needed. (b) Integration by parts — x and e^{2x} are genuinely different function types with no simplifying substitution; use $u = x$, $dv = e^{2x}dx$.

18 **[Challenging]** • **Target: Choosing between substitution and parts** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

$du = \sec x\tan x\,dx$, $v = \tan x$:
$I = \sec x\tan x - \int \sec x\tan^2 x\,dx = \sec x\tan x - \int \sec x(\sec^2 x - 1)\,dx = \sec x\tan x - \int \sec^3 x\,dx + \int \sec x\,dx$. So
$2I = \sec x\tan x + \ln|\sec x + \tan x|$, giving $I = \frac{1}{2}(\sec x\tan x + \ln|\sec x + \tan x|) + C$.

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19 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.

Total growth $= \int_0^5 t\ln(t+1)\,dt$. This does not integrate cleanly with $u = \ln(t+1)$, $dv = t\,dt$ alone; using $u = \ln(t+1)$, $dv = t\,dt$, $v = \frac{t^2}{2}$: $\left[\frac{t^2}{2}\ln(t+1)\right]_0^5 - \frac{1}{2}\int_0^5 \frac{t^2}{t+1}\,dt$. Dividing $\frac{t^2}{t+1} = t - 1 + \frac{1}{t+1}$: $\frac{1}{2}\int_0^5 (t - 1 + \frac{1}{t+1})dt = \frac{1}{2}\left[\frac{t^2}{2} - t + \ln(t+1)\right]_0^5 = \frac{1}{2}(12.5 - 5 + \ln 6) \approx 4.65$. So total growth $\approx \frac{25}{2}\ln 6 - 4.65 \approx 22.41 - 4.65 \approx 17.76$ thousand peop.

20 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Area between $y = x$ and $y = x^2$: intersect at $x =$.

Final Answer: 0 and 1. Applying Applications of Integration principles, $x = x^2 \Rightarrow x(x-1) = 0$. Therefore, the verified correct option is 0 and 1.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

State the formula for integration by parts.

a. $\int u\,dv = uv - \int v\,du$ ✓

b. $\int u\,dv = uv + \int v\,du$

c. $\int u\,dv = u'v - uv'$

d. $\int u\,dv = \frac{u}{v} - \int u'v$

Derived from the product rule: $\frac{d}{dx}(uv) = u'v + uv' \Rightarrow \int u\,dv = uv - \int v\,du$.

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22 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

Evaluate $\int xe^{2x}\,dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ ✓

b. $\frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{4}xe^{2x} - \frac{1}{4}e^{2x} + C$

$u = x, dv = e^{2x}dx \Rightarrow du = dx, v = \frac{1}{2}e^{2x}$. $\int xe^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

23 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

Evaluate $\int x\sin(x)\,dx$.

a. $-x\cos x + \sin x + C$ ✓

b. $x\cos x - \sin x + C$

c. $-x\cos x - \sin x + C$

d. $x\sin x + \cos x + C$

$u = x, dv = \sin x dx \Rightarrow du = dx, v = -\cos x$.

$uv - \int v\,du = -x\cos x - \int(-\cos x)dx = -x\cos x + \sin x + C$.

24 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int \ln(x)\,dx$.

a. $x\ln x - x + C$ ✓

b. $\frac{1}{x} + C$

c. $x\ln x + x + C$

d. $\frac{(\ln x)^2}{2} + C$

$u = \ln x, dv = dx \Rightarrow du = \frac{1}{x}dx, v = x$. $x\ln x - \int x \cdot \frac{1}{x}dx = x\ln x - x + C$.

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25 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int \arctan(x)\,dx$.

a. $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$ ✓

b. $x\arctan x - \ln(1+x^2) + C$

c. $\frac{1}{1+x^2} + C$

d. $x\arctan x + \frac{1}{2}\ln(1+x^2) + C$

$u = \arctan x, dv = dx \Rightarrow du = \frac{dx}{1+x^2}, v = x$. $x\arctan x - \int \frac{x}{1+x^2}dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C$.

26 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int x\ln(x)\,dx$.

a. $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$ ✓

b. $\frac{x^2}{2}\ln x - \frac{x^2}{2} + C$

c. $x^2\ln x - x^2 + C$

d. $\frac{x^2}{2}\ln x + \frac{x^2}{4} + C$

$u = \ln x, dv = xdx \Rightarrow du = \frac{dx}{x}, v = \frac{x^2}{2}$. $\frac{x^2}{2}\ln x - \int \frac{x}{2}dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

27 [Foundational] • Target: Repeated Integration and Tabular Method

Evaluate $\int x^2 e^x\,dx$.

a. $x^2e^x - 2xe^x + 2e^x + C$ ✓

b. $x^2e^x - 2xe^x + C$

c. $x^2e^x + 2xe^x + 2e^x + C$

d. $\frac{x^3}{3}e^x + C$

Tabular signs $(+, -, +)$: $(+)(x^2)(e^x) - (2x)(e^x) + (2)(e^x) = e^x(x^2 - 2x + 2) + C$.

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28 [Foundational] • Target: Repeated Integration and Tabular Method

Evaluate $\int x^2 \cos(x)\, dx$.

a. $x^2 \sin x + 2x \cos x - 2 \sin x + C$ ✓

b. $x^2 \sin x - 2x \cos x + 2 \sin x + C$

c. $-x^2 \cos x + 2x \sin x + C$

d. $x^2 \sin x + 2x \cos x + C$

Tabular: $D: x^2, 2x, 2, 0$; $I: \cos x, \sin x, -\cos x, -\sin x$. Pair with signs:
$x^2 \sin x - 2x(-\cos x) + 2(-\sin x) = x^2 \sin x + 2x \cos x - 2 \sin x + C$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Repeated Integration and Tabular Method

Evaluate $\int_0^1 x^2 e^{2x}\, dx$.

a. $\frac{e^2 - 1}{4}$ ✓

b. $\frac{e^2}{4}$

c. $\frac{e^2 + 1}{4}$

d. $e^2 - 1$

Tabular antiderivative: $e^{2x}\left(\frac{x^2}{2} - \frac{2x}{4} + \frac{2}{8}\right) = e^{2x}\left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{4}\right)$. At $x = 1$: $e^2(1/2 - 1/2 + 1/4) = \frac{e^2}{4}$. At $x = 0$: $1(1/4) = 1/4$. Difference $= \frac{e^2 - 1}{4}$.

30 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int e^x \cos(x)\, dx$.

a. $\frac{1}{2} e^x (\cos x + \sin x) + C$ ✓

b. $\frac{1}{2} e^x (\cos x - \sin x) + C$

c. $e^x (\sin x + \cos x) + C$

d. $\frac{1}{2} e^x \sin x + C$

Let $I = \int e^x \cos x dx$. By parts twice:
$I = e^x \sin x + e^x \cos x - I \Rightarrow 2I = e^x (\sin x + \cos x) \Rightarrow I = \frac{1}{2} e^x (\sin x + \cos x) + C$.

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31 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int_1^e \ln(x)\,dx$.

a. 1 ✓

b. $e-1$

c. e

d. 0

$[x\ln x - x]_1^e = (e\ln e - e) - (1\ln 1 - 1) = (e-e) - (0-1) = 0-(-1) = 1.$

32 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int_0^{\pi} x\sin(2x)\,dx$.

a. $-\pi/2$ ✓

b. $\pi/2$

c. 0

d. $-\pi$

Parts: $u=x$, $dv=\sin(2x)dx \Rightarrow v=-\frac{1}{2}\cos(2x)$.

$[-\frac{1}{2}x\cos(2x)]_0^{\pi} + \frac{1}{2}\int_0^{\pi}\cos(2x)dx = [-\frac{1}{2}\pi(1)-0] + [\frac{1}{4}\sin(2x)]_0^{\pi} = -\frac{\pi}{2}+0 = -\frac{\pi}{2}$.

33 [Intermediate] • Target: Repeated Integration and Tabular Method

Area between $y=e^x$ and $y=e^{-x}$ from $x=-1$ to $x=1$:

a. $2(e-1/e)$ ✓

b. $e-1/e$

c. 0

d. $e+1/e-2$

$\int_{-1}^{1}(e^x-e^{-x})dx = 2[\sinh x]_{-1}^{1} = 4\sinh 1 = 2(e-1/e)$.

34 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Area between $y=x^2$ and $y=x$ equals:

a. 1/3

b. 1/6 ✓

c. 1/2

d. 1

Intersect at 0,1. $\int_0^1 (x-x^2)dx = 1/2 - 1/3 = 1/6$.

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35 [Intermediate] • Target: The integration by parts formula

The region between $y = e^x$ and $y = 1$ from $x = 0$ to $x = 1$ has area:

a. 1

b. $e - 1$

c. e

d. $e - 2$ ✓

$\int_0^1 (e^x - 1)dx = [e^x - x]_0^1 = (e - 1) - 1 = e - 2.$

36 [Intermediate] • Target: Repeated and definite applications

Tabular integration (repeated parts): $\int x^3 e^x \, dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3 ✓

x^3 differentiates to 0 in 3 steps.

37 [Intermediate] • Target: Choosing u and dv wisely (LIATE)

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)| \, dx$ ✓

b. $\int_a^b f(x) \, dx$ only when $f \geq 0$

c. $f(b) - f(a)$

d. $\int_a^b f(x)^2 \, dx$

Always use absolute value for area.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The recurring-integral trick • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1}x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2}x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

a. $5\pi/16$

b. $3\pi/16$ ✓

c. $5\pi/32$

d. $15\pi/48$

Wallis: $\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi\cdot 3}{6\cdot 4\cdot 2\cdot 2} = \frac{15\pi}{96} = \frac{5\pi}{32}$. Actually Wallis: $\int_0^{\pi/2}\sin^{2n}x\,dx = \frac{(2n-1)!!}{(2n)!!}\frac{\pi}{2}$. For $n=3$: $\frac{5\cdot 3\cdot 1}{6\cdot 4\cdot 2}\frac{\pi}{2} = \frac{15}{48}\frac{\pi}{2} = \frac{15\pi}{96} = \frac{5\pi}{32}$.

39 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_\alpha^\beta r\,d\theta$

b. $\int_\alpha^\beta |dr/d\theta|\,d\theta$

c. $\int_\alpha^\beta r^2\,d\theta$

d. $\int_\alpha^\beta \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$ ✓

Applying Applications of Integration principles, Polar arc length formula. Therefore, the verified correct option is $\int_\alpha^\beta \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$.

40 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Trig substitution $\int \sqrt{4 - x^2}\,dx$ with $x = 2\sin\theta$ gives:

a. $\arcsin(x/2) + C$

b. $\frac{x\sqrt{4-x^2}}{2} + 2\arcsin(x/2) + C$ ✓

c. $\sqrt{4 - x^2} + C$

d. $-\sqrt{4 - x^2} + C$

Result of trig sub for $\int \sqrt{a^2 - x^2}\,dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\arcsin(x/a) + C$. With $a = 2$: $\frac{x\sqrt{4-x^2}}{2} + 2\arcsin(x/2) + C$.

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41 [Challenging] • Target: Integration by parts with inverse trig functions • Cross-Lesson: 6.01 Related Rates

Centroid $\bar{x}$ of region R with area A:

a. $\frac{a+b}{2}$

b. $\frac{1}{A}\int_a^b f(x)\,dx$

c. $\frac{1}{A}\int_a^b x[f(x)-g(x)]\,dx$ ✓

d. $A/2$

Applying Applications of Integration principles, Centroid formula. Therefore, the verified correct option is $\frac{1}{A}\int_a^b x[f(x)-g(x)]\,dx$.

42 [Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 6.02 Optimization and Curve Sketching

Arc length formula: $L =$

a. $\int_a^b f(x)dx$

b. $\int_a^b \sqrt{1+[f'(x)]^2}\,dx$ ✓

c. $\int_a^b |f'(x)|dx$

d. $\int_a^b [1+f'(x)]dx$

Applying Applications of Integration principles, Arc length formula. Therefore, the verified correct option is $\int_a^b \sqrt{1+[f'(x)]^2}\,dx$.

43 [Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 7.01 Integration by Substitution

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1}x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2}x\,dx$. For $n=4$:

a. $-\sin^3 x\cos x/4 + 3/4\int \sin^2 x\,dx$ ✓

b. $-\sin^3 x\cos x$

c. $\sin^4 x/4$

d. $-\cos x/4$

Applying Integrals principles, Standard reduction with $n=4$. Therefore, the verified correct option is $-\sin^3 x\cos x/4 + 3/4\int \sin^2 x\,dx$.

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44 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 8.01 Area Between Curves

Shell method: volume when shell radius $= x$, height $= f(x)$:

a. $2\pi\int_a^b xf(x)dx$ ✓

b. $\int_a^b f(x)dx$

c. $\pi\int_a^b x^2 f(x)dx$

d. $\pi\int_a^b f(x)^2 dx$

Applying Applications of Integration principles, $V = 2\pi\int_a^b xf(x)dx$. Therefore, the verified correct option is $2\pi\int_a^b xf(x)dx$.

45 [Challenging] • Target: Standard Products (Polynomial × Exponential / Trig) • Cross-Lesson: 8.02 Volumes of Revolution

Improper integral $\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx$:

a. -4 ✓

b. diverges

c. 0

d. -2

$u = \ln x$, $v = 2\sqrt{x}$: IBP gives $[2\sqrt{x}\ln x]_0^1 - \int_0^1 2/\sqrt{x}\cdot 1/x\cdot(-1)$... Actually: $\int_0^1 x^{-1/2}\ln x\,dx = -4$.

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